

# Variable Length Subnet Mask

Course Code: CSC 3116

Course Title: Computer Networks



**Dept. of Computer Science**  
**Faculty of Science and Technology**

|                     |                                                                                 |                 |   |                  |  |
|---------------------|---------------------------------------------------------------------------------|-----------------|---|------------------|--|
| <b>Lecturer No:</b> | Lab 4                                                                           | <b>Week No:</b> | 2 | <b>Semester:</b> |  |
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# Lecture Outline



1. Introduction
2. VLSM
  - Steps of VLSM
  - Example 1
  - Example 2
3. Homework

# Introduction



## Problems of Large Network

- A large network is difficult to manage
- Less security as each host of the network can reach all other hosts.
- Huge broadcast domain, thereby large bandwidth consumption
- Waste of unused IP addresses

# Introduction....



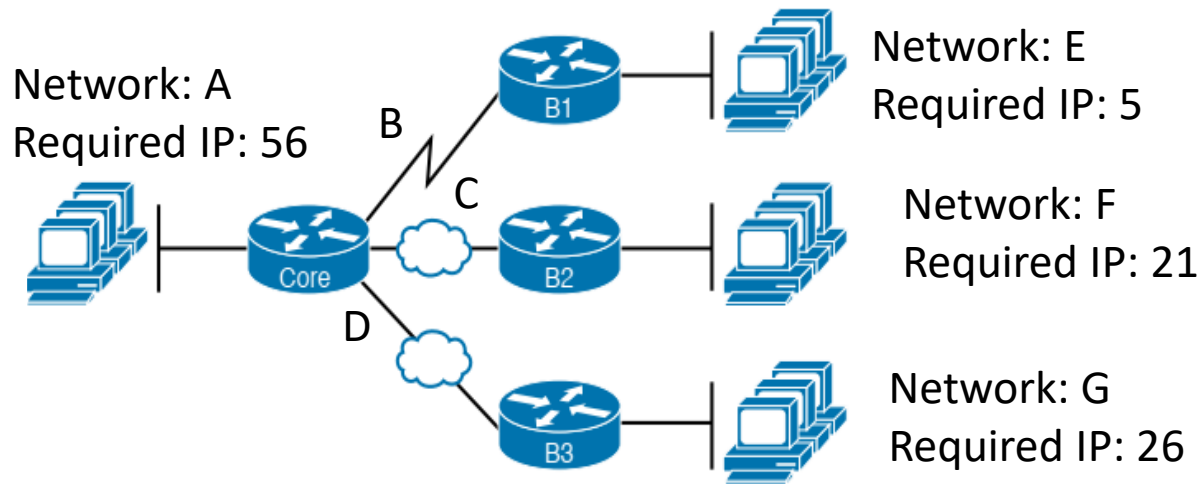
## How to overcome the problems?

Dividing a large network into several smaller networks,  
called subdivided networks or subnets.

# Introduction....



For classful addressing



| Network | No. of IP addresses allocated |
|---------|-------------------------------|
| A       | 256                           |
| B       | 256                           |
| C       | 256                           |
| D       | 256                           |
| E       | 256                           |
| F       | 256                           |
| G       | 256                           |

# Introduction



- Required IPs:  $56+5+21+26+4+4+4=120$
- **Classful address**
  - No. o allocated IPs:  $256 \times 7 = 1792$  IP
  - Percentage of unused IP  $= \frac{(1792-120) \times 100}{1792} = 93\%$  (approx.)

# Introduction



Is there any way to further reduce the  
unused IP addresses?

**Yup!**

Instead of giving the same number of IP addresses to all subnets,  
allocate different number of IP addresses depending  
on each subnet's needs.

# VLSM



- Variable Length Subnet Mask (VLSM)
- Instead of allocating the same number of IP addresses to all networks, the number of IP addresses allocated to network depends on the network's need.
- Different network is provided with different number of IPs



# VLSM

## Steps of VLSM



- *Step 1:* Write the networks' names and IP requirements in descending order of IP requirements.  
A (56), G(26), F(21), E(5), B(4), C(4), D(4)
- *Step 2:* Find the number of host bits needed to satisfy the IP requirement of each subnet?
- *Step 3:* Based on the number of host bits, find the subnet mask for each subnet.
- *Step 4:* Allocate IPs to each subnet starting from the beginning of the IP block. Allocate IPs to subnet sequentially according to the sorted network sequence found in Step 1.

# VLSM....

## Example 1



11111111. 11111111. 11111111. 11000000

26

6

| Subnet | No. of IPs required | How many bits to borrow | No. of allocated IPs | No. of host bits<br>No. of net bits | Subnet mask     | Allocated IP range                 |
|--------|---------------------|-------------------------|----------------------|-------------------------------------|-----------------|------------------------------------|
| A      | 56                  | $2^6 > 56 > 2^5$        | 64                   | $x = 6,$<br>$y = 32 - 6 = 26$       | 255.255.255.192 | 192.168.5.0-<br>192.168.5.63/26    |
| G      | 26                  | $2^5 > 26 > 2^4$        | 32                   | $x = 5,$<br>$y = 32 - 5 = 27$       | 255.255.255.224 | 192.168.5.64-<br>192.168.5.95/27   |
| F      | 21                  | $2^5 > 21 > 2^4$        | 32                   | $x = 5,$<br>$y = 32 - 5 = 27$       | 255.255.255.224 | 192.168.5.96-<br>192.168.5.127/27  |
| E      | 5                   | $2^3 > 5 > 2^2$         | 8                    | $x = 3,$<br>$y = 32 - 3 = 29$       | 255.255.255.248 | 192.168.5.128-<br>192.168.5.135/29 |
| B      | 4                   | $2^2 = 4$               | 4                    | $x = 2,$<br>$y = 32 - 2 = 30$       | 255.255.255.252 | 192.168.5.136-<br>192.168.5.139/30 |
| C      | 4                   | $2^2 = 4$               | 4                    | $x = 2,$<br>$y = 32 - 2 = 30$       | 255.255.255.252 | 192.168.5.140-<br>192.168.5.143/30 |
| D      | 4                   | $2^2 = 4$               | 4                    | $x = 2,$<br>$y = 32 - 2 = 30$       | 255.255.255.252 | 192.168.5.144-<br>192.168.5.147/30 |

# VLSM.....

## Example 1



## VLSM

No. of allocated IPs:  $64 + 32 + 32 + 8 + 4 + 4 + 4 = 148$  IP

Percentage of unused IP =  $\frac{(148 - 120) \times 100}{148} = 19\%$  (approx.)

Percentage of unused IP address (for the given network)

| Classful Addressing | VLSM |
|---------------------|------|
| 93%                 | 19%  |

# VLSM.....



Is it possible to further reduce the unused IP?

Spend your time to propose a new subnetting technique.

# VLSM....

## Example 2



- ❖ Suppose that we have three networks: A, B and C with IP requirements 50, 4 and 28. If You are given an IP block 130.3.0.0, allocate IPs performing subnetting.

| Subnet | No. of IPs required | How many bits to borrow | No. of allocated IPs | No. of host bits<br>No. of net bits | Subnet mask     | Allocated IP range           |
|--------|---------------------|-------------------------|----------------------|-------------------------------------|-----------------|------------------------------|
| A      | 50                  | $2^6 > 50 > 2^5$        | 64                   | $x = 6,$<br>$y = 32 - 6 = 26$       | 255.255.255.192 | 130.3.0.0-<br>130.3.0.63/26  |
| C      | 28                  | $2^5 > 28 > 2^4$        | 32                   | $x = 5,$<br>$y = 32 - 5 = 27$       | 255.255.255.224 | 130.3.0.64-<br>130.3.0.95/27 |
| B      | 4                   | $2^2 = 4$               | 4                    | $x = 2,$<br>$y = 32 - 2 = 30$       | 255.255.255.252 | 130.3.0.96-<br>130.3.0.99/30 |

# Lab Task



1. Suppose that we have three networks: A, B and C with IP requirements 500, 4000 and 208, respectively. If You are given an IP block 130.3.0.0, allocate IPs performing subnetting.
2. Suppose that we have six networks: A, B, C, D, E and F with IP requirements 120, 4, 9, 4, 32, and 7, respectively. If You are given an IP block 210.3.0.0, allocate IPs performing subnetting.

# References



[No reference]



## Recommended Books

1. **Official Cert Guide CCNA 200-301 , vol. 1**, *W. Odom*, Cisco Press, First Edition, 2019, USA.
2. **CCNA Routing and Switching**, *T. Lammle*, John Wily & Sons, Second Edition, 2016, USA.